THE SYNCHRONISED TREE · 1 / 11

Four inputs. One sum.

How do values from different threads meet?

> PyTorch chooses its implementation. This is a separate teaching algorithm.



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THE SYNCHRONISED TREE · 2 / 11

Write into a shared workspace.

Shared memory lets threads in one block exchange values.

> Snapshot after one input write. Unwritten is not zero.



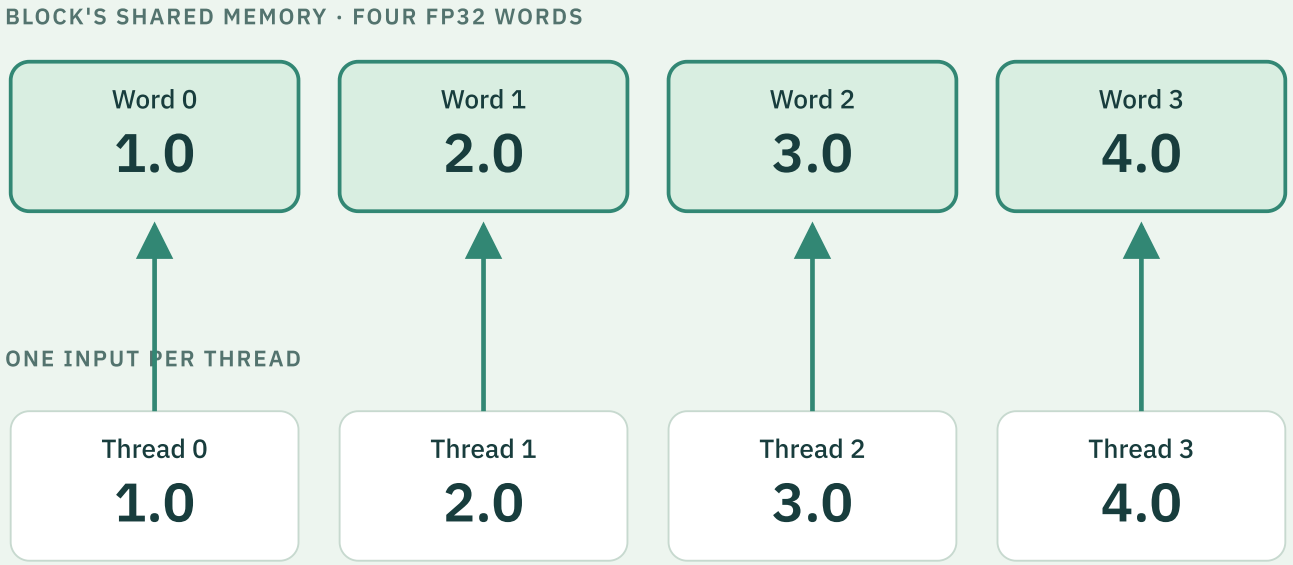
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THE SYNCHRONISED TREE · 3 / 11

Each thread writes its input.

The picture shows writes; the program still needs ordering.

> Inspection order is not a promised thread execution order.



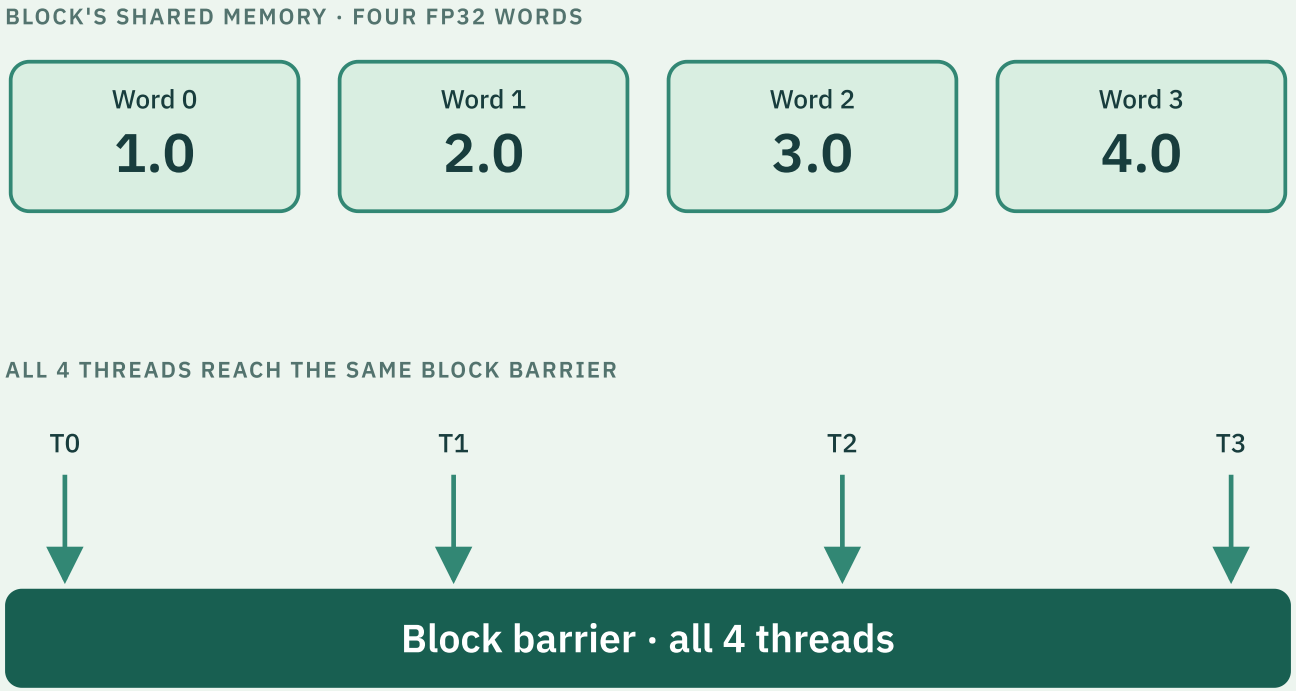
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THE SYNCHRONISED TREE · 4 / 11

Wait before reading a partner.

Every thread in the block reaches the barrier.

> The initial barrier orders input writes before partner reads.



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THE SYNCHRONISED TREE · 5 / 11

Combine partners two words apart.

Each partial sum contains a different pair of inputs.

> Partner distance 2 · the partial sums contain disjoint input sets.



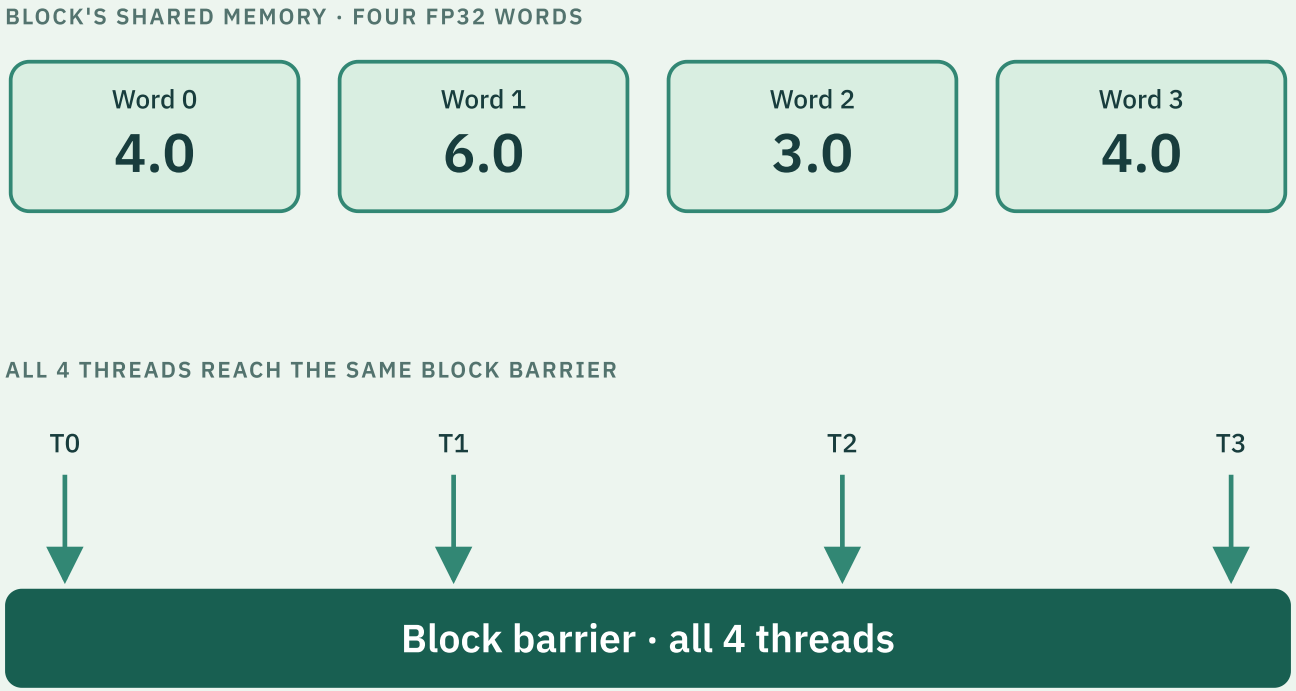
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THE SYNCHRONISED TREE · 6 / 11

Wait again before the next stage.

Only two threads added. All four still reach the barrier.

> Threads without an addition still reach the block barrier.



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THE SYNCHRONISED TREE · 7 / 11

Combine the two partial sums.

Every original input contributes exactly once.

> Dependency arrows are not a hardware timing diagram.



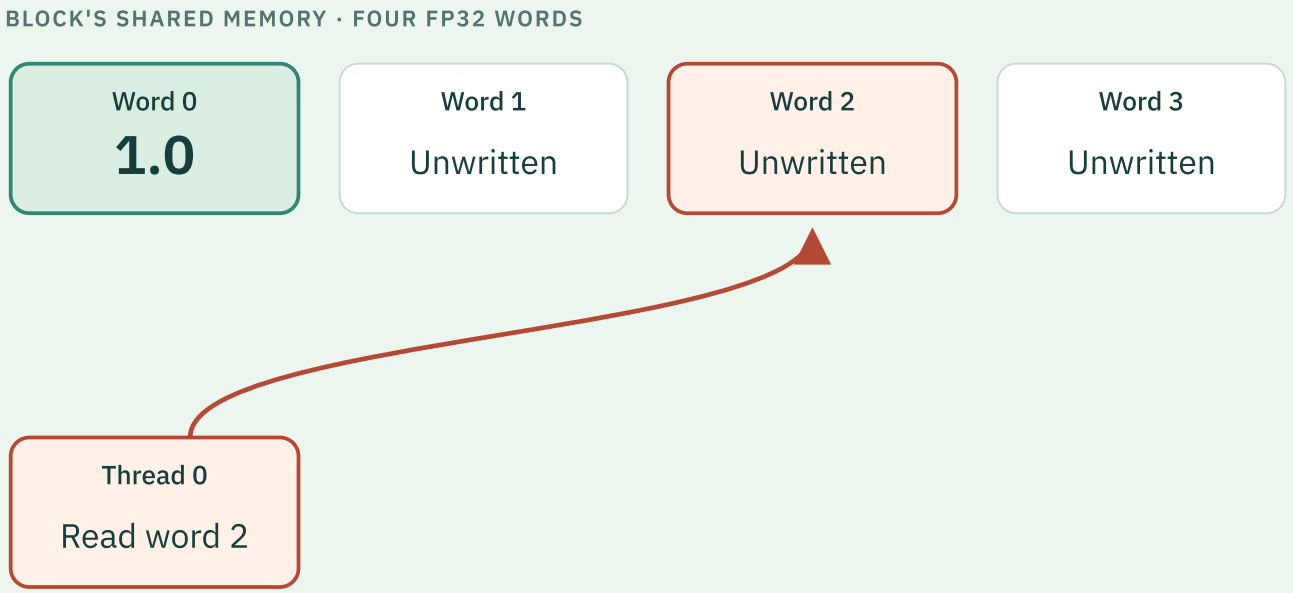
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THE MISSING DEPENDENCY · 8 / 11

What if the ordering is missing?

One diagnostic interleaving: a read arrives before its write.

> Do not invent a numeric result for a data race.



Unwritten partner read · no numeric result

One diagnostic interleaving; other orders are possible.

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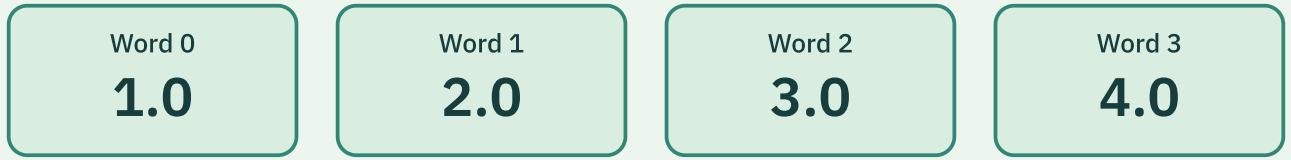
THE MISSING DEPENDENCY · 9 / 11

A later write cannot repair the read.

The required write must precede the read that uses it.

> Correctness requires ordering before the read.

BLOCK'S SHARED MEMORY · FOUR FP32 WORDS



The earlier read is still invalid.

Writing word 2 later cannot repair it.

A barrier only after the read is too late.

Racy program → no numeric result assigned

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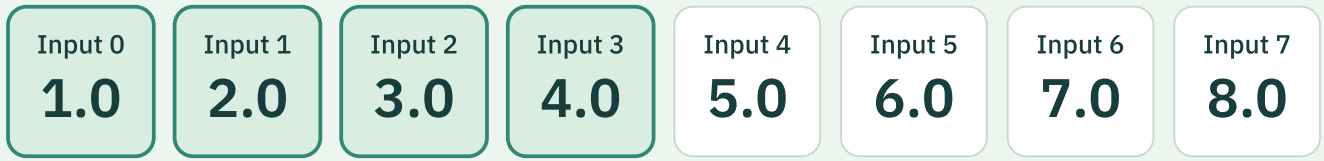
TRY A NEW CASE · 10 / 11

Pause: extend the block to eight inputs.

Only four threads add in the first stage.

> Pause the video before checking the next slide.

EIGHT INPUTS · ONE BLOCK · PARTNER DISTANCE 4



Thread 1 reads word 5. Which write must come first?

ONLY THREADS 0–3 ADD IN THE FIRST STAGE

T0 T1 T2 T3 T4 T5 T6 T7

Block barrier · how many threads?

Pause and explain both answers.

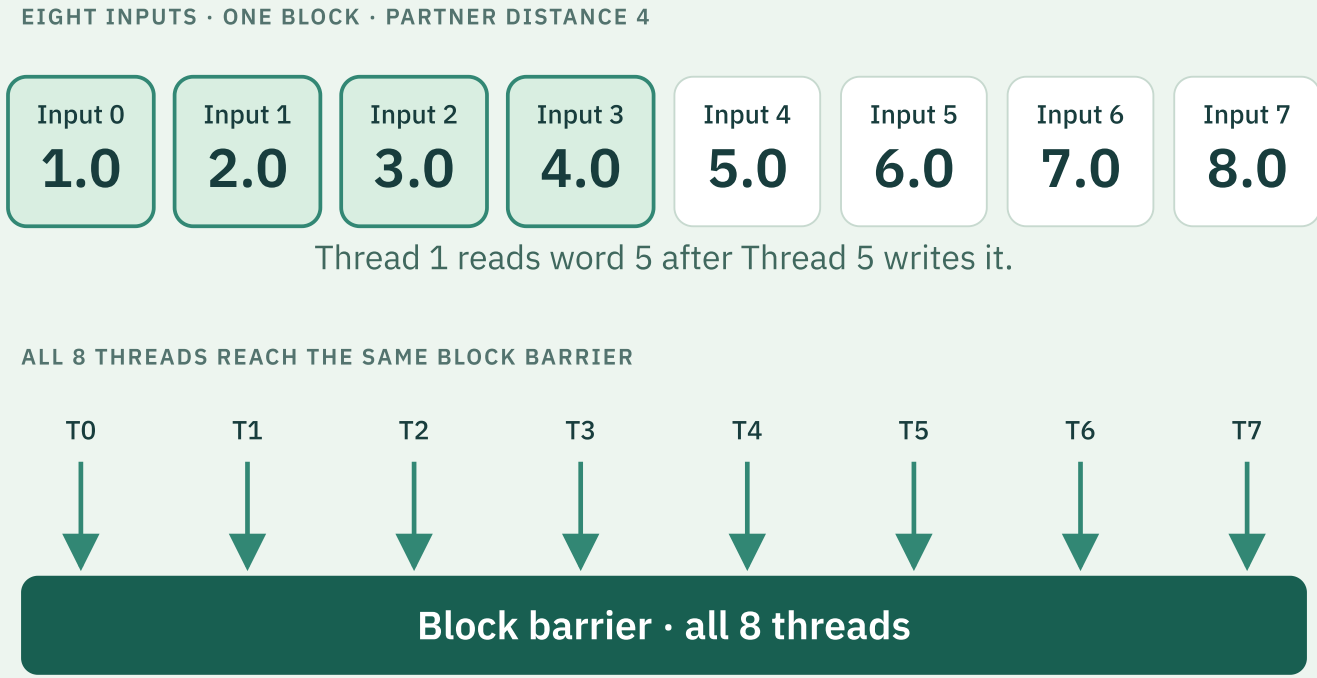
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TRY A NEW CASE · 11 / 11

All eight threads reach the barrier.

Thread 5 writes before thread 1 reads word 5.

> Return to Chapter 9 → extend the synchronised tree to eight inputs.



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